

DRAGOS PATCA

Mechatronics Engineering Student · Artist Manager · Freelance Engineer

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PROFILE

Hands-on mechatronics engineering student at SRH University Berlin, with a background in CNC manufacturing, electronics, and autonomous systems design. Proven ability to take projects from concept in SolidWorks through to real-world assembly and production. Simultaneously manages and grows Riveria Band's digital presence, reaching 20K+ views per post through data-driven social media strategy. Quadrilingual (Romanian / English C1 / German B2 / Italian A2) with a Cambridge certificate and strong cross-cultural communication skills.

EDUCATION

B.Eng. Mechatronics Engineering

2025 – Present

SRH University Berlin · Berlin, Germany

- Focus: autonomous systems, robotics, CAD, embedded electronics, software
- Key projects: Follice Box (autonomous follow-me cooler), Autonomous Polishing Assembly

Technical High School Diploma (Abitur equivalent) — C++ Advanced

2021 – 2025

Colegiul Tehnic "INFOEL" · Bistrița, Romania

- Final exam (equivalent to Abitur) with advanced C++ specialisation
- Teaching certificate obtained during studies
- Participated in robotics competitions throughout high school

EXPERIENCE

Artist Manager & Social Media Strategist — Riveria Band

2023 – Present

Self-employed · Remote / Berlin

- Manage full social media strategy, content production, and event coordination for the band
- Grew Instagram to 2,500 followers with 20,000+ views per post through targeted promotion
- Handle visual identity, audience development, and promotional partnerships

Freelance Bicycle Repair & Resale

2022 – 2024

Self-employed · Bistrița, Romania

- Sourced, repaired, and resold second-hand bicycles, applying hands-on mechanical skills
- Managed customer relations, pricing, and logistics independently

Voluntary Participant — Logimat Trade Fair

2025

Stuttgart, Germany

- Attended Europe's leading intralogistics and engineering trade fair as a student delegate
- Networked with industry professionals and explored cutting-edge automation solutions

Volunteer — Europe Direct & Cultural Events

2021 – 2025

Bistrița, Romania · 4 years continuous

- Active volunteer across multiple events: Târgul de Purici, Artcast Teleki Festival, Europe Direct programmes
- Responsibilities included event setup, visitor engagement, and team coordination

ENGINEERING PROJECTS

Follice Box — Autonomous Follow-Me Cooler2024 – Present

- Designed full system in SolidWorks: 50L insulated chassis, 24V differential drive, obstacle detection
- Engineered for 1.2 m/s top speed, 10° slope capability, ~3h runtime on Li-ion 7S battery
- Selected hardware: ESP32, Cytron MDDS30 motor driver, ToF VL53L1X sensor, magnetic encoders

Autonomous Polishing Assembly — CNC Project2024

- Designed and built a full polishing machine using Bosch profiles, Motovario motor, custom CNC parts
- Manufactured custom 2-piece motor bracket and rotating magnetic disc (8 magnets) on CNC
- Integrated dual latching buttons and rotary emergency stop in a custom electrical control panel
- Machine currently in active production use at the client company

Python / Pygame Game Development2024

- Developed an interactive game in Python using the Pygame library
- Deployed as a browser-playable web build via pygbag

SKILLS & COMPETENCIES

Engineering & Manufacturing

- CAD: SolidWorks, Autodesk Inventor Pro(3D modelling, assemblies, 2D drawings, BOM) / Fusion 360
- CNC machining, metalworking, welding, brazing, woodworking — hands-on production experience
- Electronics: circuit design, ESP32, Arduino, motor drivers, BMS, encoders, ToF sensors

Programming

- C++ / C / C# (Advanced — certified), Python (Pygame), basic embedded firmware (ESP32/Arduino)

Creative & Management

- Social media strategy, content creation, audience growth, event coordination
- Video production and editing, visual identity management

CERTIFICATIONS & ACHIEVEMENTS

- Cambridge English C1 Advanced Certificate
- Teaching Certificate — obtained during technical high school
- C++ Advanced — Final Exam, Colegiul Tehnic INFOEL
- European Driving Licence — Category B (cars & vans/minibuses)
- Robotics Competition Participant — high school level
- European Cup Competitor — Downhill Mountain Biking
- Musician — 60+ live concerts & 5+ own songs

LANGUAGES

Romanian: Native	English: C1 — Cambridge Certified	German: B2	
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INTERESTS

Autonomous robotics & embedded systems · Mountain biking (Downhill, competitive) · Music production & artist management · CNC manufacturing · Open-source hardware